and Siddarth, 2023). This could also undermine the authentication of evidence by making it easier for people to reject genuine audiovisual evidence of wrongdoing as "fake" (Bond, 2023; Citron and Chesney, 2019; Pawelec, 2022; Pfefferkorn, 2020).

Evaluating *misinformation* harms comprehensively requires measuring different concepts at each of the three layers of a sociotechnical approach to evaluation. To make multifaceted harms such as misinformation tractable, it can be helpful first to identify concepts that relate to or constitute the overall harm. Such concepts sit between the latent construct (e.g. "misinformation") and the concrete measure (e.g. FID scores). An example is "factuality". Factuality is conceptually narrower than the high-level concepts of harm, but it is broader than the specific metrics that may be used to measure it (e.g. FID scores). Adding this intermediate step of identifying narrower concepts that compose a harm can help build the bridge between latent constructs and concrete metrics. It can also indicate where an evaluation may be able to shed light on multiple risk areas.

A.3.1. Capability

Operationalisation: Factuality of AI system outputs. The likelihood of an AI system generating misinformation can be assessed via benchmarks and other tests at the capability layer. Different methods have been proposed to assess and improve the factual accuracy of outputs from generative AI systems. Due to the widespread availability of textual data, most focus on the text modality; in contrast, evaluations of factuality for image, audio, and video modalities are sparse.

Several approaches involve automatic *fact verification* – i.e. fact-checking AI system outputs against existing knowledge sources (e.g. Wikipedia articles or databases of factual statements such as WikiData) (Lee et al., 2023c; OpenAI, 2023b). Others focus on *verifiability*, testing whether statements generated by an AI system can be attributed to reliable external sources (FActScore, Min et al. (2023); RARR, Gao et al. (2023); Liu et al. (2023a)). Wiki-FACTOR and News-FACTOR (Muhlgay et al., 2023) as well as FactualityPrompt (Lee et al., 2023c) test an AI system's factual knowledge, i.e. its propensity to generate factual statements from a corpus versus incorrect ones, while TruthfulQA measures the likelihood of its generating false answers learned from imitating human texts in response to open-ended questions (Lin et al., 2022). Recent work in this area focuses on improving factuality by training AI systems to automatically detect false claims, multimodal misinformation, and conspiracy theories (Papadopoulos et al. (2023); HaluEval, Li et al. (2023a); HaDes, Liu et al. (2022); Russo et al. (2023)).

However, these benchmarks capture a limited operationalisation of "truthfulness". In practice, a more nuanced and contextual evaluation of factuality often requires human expertise. Some evaluations involve testing the AI system's propensity to generate false and potentially harmful narratives (Brewster et al., 2023; Center for Countering Digital Hate, 2023b). Most human annotation focuses on evaluating whether AI system outputs can be supported by references. There are two main approaches to this human annotation. One is to instruct annotators to fact-check the output against existing knowledge bases – by performing a Google or Wikipedia search, for example (Maynez et al., 2020) – or to rate the degree of confabulation within a range (Ji et al., 2023). The other approach is to measure the proportion of AI system responses that can be attributed to identified sources (AIS) (Rashkin et al., 2022). However, these approaches do not account for source quality and trustworthiness.

Identifying and evaluating outputs for truthfulness present new challenges for audiovisual modalities, as this type of content may not be as easily verifiable against existing public knowledge bases. Some knowledge bases exist, such as Fakeddit (Nakamura et al., 2020), Factify (Mishra et al., 2022), and Fake2M (Lu et al., 2023). Several datasets also exist to aid the detection of audio and video deepfakes (APTLY; FakeAVCeleb, Khalid et al. (2022)). Even so, the lack of large, up-to-date datasets for multimodal verification remains a major

roadblock. For audio and image generation, as well as multimodal summarisation, alignment metrics e.g. CLIPScores (Hessel et al., 2022), CLAPScores (Elizalde et al., 2022; Radford et al., 2021), and CLIPBERTScore (Wan and Bansal, 2022) can help evaluate how well an AI system represents the relationship between audio or images and text (Otani et al., 2023), though these metrics are quite general and not suitable for the detection of AI-generated audiovisual content.

Manually checking the factual accuracy of AI system outputs is a laborious task that hinges on raters' prior knowledge and fact-checking abilities, and on the availability of observable evidence against which to verify information. This is all the more complicated in cases where notions of "truth" are debated (such as political matters) or on topics where consensual knowledge is emerging and subject to change (e.g. Covid-19) (Evans et al., 2021; Kasirzadeh and Gabriel, 2023). For that reason, factuality evaluations can quickly become outdated. While it is possible to train an AI system to not output content that contradicts existing scientific facts – assertions of flat earth theory, for instance – this is more challenging for novel content that is not already present in some existing data.

Operationalisation: Credibility of AI system outputs. Across all modalities, one key challenge of existing approaches to evaluating misinformation harms is that content factuality is not a sufficient proxy for determining whether a piece of content has the potential to misinform.[17] Some fictitious or factually incorrect content may be harmless because it defies common sense and is therefore unlikely to be believed at face value (e.g. a video of a cat scuba diving). Conversely, some types of outputs have a higher chance of deceiving or misinforming audiences if they are perceived as credible or believable, even if they are not factually accurate (Klein et al., 2023). One way to evaluate the credibility of multimodal outputs at the AI system layer is to measure their perceptual quality or realism. Automated tools assess quality and fidelity scores (e.g. FID scores, Inception scores), which measure how closely generated images and audio resemble real reference outputs (Heusel et al., 2018; Salimans et al., 2016). However, fidelity scores in particular have been criticised for correlating poorly with human perceptions of quality (Saharia et al., 2022).

Human evaluations also tend to focus on perceptions of image and speech quality (e.g. Mean Opinion Scores of the realism and perceptual quality of synthetic outputs compared to natural images) (Kong et al., 2021; Otani et al., 2023). Other benchmarks specifically test human ability to distinguish real and AI-generated images – an example of a benchmark that targets usability layer outcomes (HPBench, Lu et al. (2023)). Common datasets and benchmarks used for qualitative benchmarking of image generation AI systems include Drawbench, MSCOCO and PartiPrompts (Lin et al., 2014; Saharia et al., 2022; Yu et al., 2022). Perceptions of realism and credibility can be influenced by a number of factors, however, including an individual's prior beliefs, knowledge, and trust in the source of the message, and the context in which they see the message. As a result, the credibility of a particular AI system output and how likely it is to deceive end users will differ based on user demographics and AI use cases. Disentangling these mechanisms requires targeted user testing at the human–AI interaction layer.

Operationalisation: Societal relevance of AI system outputs. Compelling AI-generated synthetic content may run a greater risk of causing harm in politically salient contexts or during particular periods, such as prior to an election. AI-generated videos of police brutality or protests, for example, have the potential to sow division and shape public opinion about political events (Taylor, 2023). Developing evaluation metrics for "social relevance" or "newsworthiness" of AI system outputs at this layer is therefore an important step toward evaluating potential downstream harms. Operationalising these

[17] A common form of audiovisual misinformation, for example, is decontextualisation – that is, using real images to provide evidence in support of a false claim (Weikmann and Lecheler, 2022).

terms into concrete metrics is often difficult in practice, as these are complex concepts whose interpretation differs based on locale. Expert fact-checkers, journalists, and misinformation experts have the necessary expertise to establish a set of criteria to guide the evaluation of multimodal AI system outputs that pose a high misinformation risk.[18] These efforts could consider factors like the domain of the content generated (e.g. legal, medical, or news-related) or the likelihood that a specific piece of content will be used to support factual statements or claims. Expert evaluation might also assess the emotional salience of AI system outputs (e.g. whether it evokes fear, anger, or moral outrage), as this is widely considered a key characteristic of persuasive misinformation (Crockett, 2017; Han et al., 2020). However, the gold standard to assess this would be to study the behavioural reactions of people exposed to these types of content.

A.3.2. Human interaction

Operationalisation: Deception. A key concept to measure at this layer is whether AI system outputs actually deceive end users (Park et al., 2023b). Recent efforts in this area focus on testing people's ability to distinguish different types of outputs as synthetic or human-generated and their ability to identify whether they contain misinformation. Several studies find that people are more easily deceived by AI-generated than human-generated misinformation (Clark et al., 2021; Groh et al., 2022; Kreps et al., 2022; Nightingale and Farid, 2022; Spitale et al., 2023). Others investigate the deceptive qualities of AI-generated content. Using qualitative and linguistic analysis, Zhou et al. (2023), for example, find that AI-generated and human-generated misinformation present significant linguistic differences, and that AI-generated misinformation is adept at "mimicking the attributes of existing information assessment guidelines", including credibility and comprehensiveness. Generative AI systems that mimic speech patterns of users are also more often believed (Chiesurin et al., 2023). Finally, several studies investigate cues of deceptive behaviour, such as facial expressions and gestures, in multimodal audiovisual content (e.g. Soldner et al. (2019); Zhang et al. (2020)).

Operationalisation: Persuasion. Other evaluations at this layer might focus on testing whether interacting with an AI system influences a person's beliefs, attitudes, and behaviour (DeVerna et al., 2023). This can be tested by measuring whether exposure to synthetic outputs convinces people to change their political views, or leads them to develop false or inaccurate beliefs about external reality (Dehnert and Mongeau, 2022). For example, Bai et al. (2023a) find AI-generated texts to be more persuasive than human-generated ones in shifting people's attitudes on a range of policy issues. Goldstein et al. (2023) test the persuasiveness of AI-generated political propaganda through a series of survey experiments.

Human-centred evaluations and experiments could also shine light on important underlying mechanisms and contextual drivers of misinformation, such as the extent to which different users are persuaded by different types of synthetic outputs. Prior studies identify several cognitive, social, and demographic factors that influence people's receptivity to misinformation and propensity to develop false beliefs – factors such as partisanship, cognitive biases, or trust in information sources (for a review, see Ecker et al. (2022)). For example Pennycook et al. (2018) demonstrate that repeated exposure increases people's belief in misinformation. Lovato et al. (2023) find that people are better at identifying deepfake videos where synthetic personas match their demographics. Likewise, psychological studies show that emotional appeals increase the persuasiveness of misinformation and its impact on misperceptions (Lee et al., 2023a; Martel et al., 2020; Tannenbaum et al., 2015).

Misinformation has the potential to deceive individuals not only by instilling falsehoods but by leading them to disbelieve true evidence and information (Rini, 2021). It is therefore critical to test how interacting with an AI

[18] Similar efforts have been undertaken for automatic claim detection (Konstantinovskiy et al., 2021) and automatic detection of potentially malicious threads on the DarkWeb (Jin et al., 2023).